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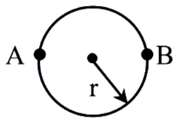
Subjects : Physics **Distance and Displacement**

Total Marks : 52 **(Solutions)**

Physics - Section A (MCQ)

1. A sportsman runs around a circular track of radius r such that he traverses the path $ABAB$. The distance travelled and displacement, respectively, are



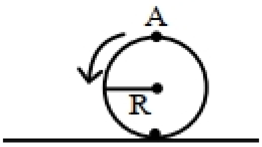
A) $2\mathrm{r}, 3\pi\mathrm{r}$ B) $3\pi r, \pi r$ C) $\pi r, 3r$ D) $3\pi r, 2r$

Solution : (Correct Answer: D)

Displacement $= 2r$

Distance $= 2\pi r + \pi r = 3\pi r$

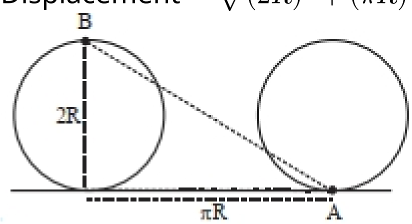
2. A disc is rolling without slipping on a surface. The radius of the disc is R. At $t = 0$, the top most point on the disc is A as shown in figure. When the disc completes half of its rotation, the displacement of point A from its initial position is



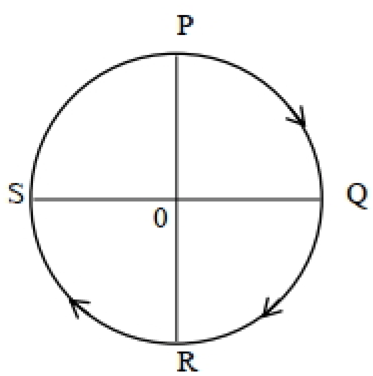
A) $R\sqrt{(\pi^2+4)}$ B) $R\sqrt{(\pi^2+1)}$ C) $2R$ D) $2R\sqrt{(1+4\pi^2)}$

Solution : (Correct Answer: A)

Displacement $= \sqrt{(2R)^2 + (\pi R)^2} = R\sqrt{4+\pi^2}$



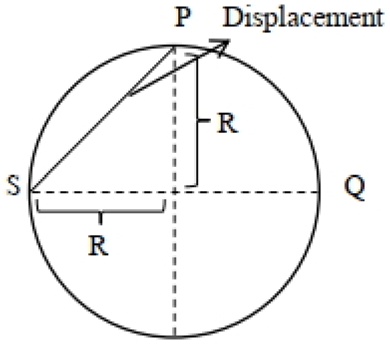
3. A cyclist starts from the point P of a circular ground of radius $2\ \mathrm{km}$ and travels along its circumference to the point S. The displacement of a cyclist is _______.



A) $6\ \mathrm{km}$ B) $\sqrt{8}\ \mathrm{km}$ C) $4\ \mathrm{km}$ D) $8\ \mathrm{km}$

Solution : (Correct Answer: B)

$\therefore$ Displacement $= \mathrm{R}\sqrt{2} = 2\sqrt{2} = \sqrt{8}$ km



4. A body starts moving from rest with constant acceleration covers displacement S_1 in first $(p-1)$ seconds and S_2 in first p seconds. The displacement $S_1 + S_2$ will be made in time :

A) $(2\mathrm{p}+1)\mathrm{s}$ **B)** $\sqrt{(2\mathrm{p}^2-2\mathrm{p}+1)\,\mathrm{s}}$ **C)** $(2\mathrm{p}-1)\mathrm{s}$ **D)** $(2p^2-2p+1)\,s$

Solution : (Correct Answer: B)

S_1 in first $(p-1)\mathrm{sec}$

S_2 in first p sec

$\mathrm{S}_1 = \frac{1}{2}\mathrm{a}(\mathrm{p}-1)^2$

$\mathrm{S}_2 = \frac{1}{2}\mathrm{a}(\mathrm{p})^2$

$\mathrm{S}_1 + \mathrm{S}_2 = \frac{1}{2}\mathrm{at}^2$

$(\mathrm{p}-1)^2 + \mathrm{p}^2 = \mathrm{t}^2$

$\mathrm{t} = \sqrt{2\mathrm{p}^2 + 1 - 2\mathrm{p}}$

5. A particle is moving along a circle such that it completes one revolution in 40 seconds. In 2 minutes 20 seconds, the ratio $\frac{|\text{ displacement }|}{\text{distance}}$ is ....

A) 0 **B)** $\frac{1}{7}$ **C)** $\frac{2}{7}$ **D)** $\frac{1}{11}$

Solution : (Correct Answer: D)

$T = 40\,s$

If $t = 2$ minute 20 second

$= 2 \times 60 + 20$

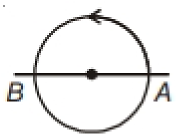
$= 140\,s$

So, it has completed $3\frac{1}{2}$ revolution.

Distance travelled $= 3 \times 2\pi R + \pi R = 7\pi R$

Displacement $= 2R$

$\frac{|\text{ Displacement}|}{\text{Distance}} = \frac{2R}{7\pi R} = \frac{2}{7\times\frac{22}{7}} = \frac{1}{11}$



6. Consider the motion of the tip of the second hand of a clock. In one minute (R be the length of second hand), its ......

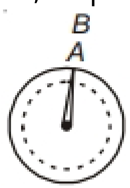
A) Displacement is $2\pi R$ **B)** Distance covered is $2R$

C) Displacement is zero **D)** Distance covered is zero

Solution : (Correct Answer: C)

The second hand of the clock in minute covers an angle of 360° and the initial and final positions are same.

So, Displacement $= 0$



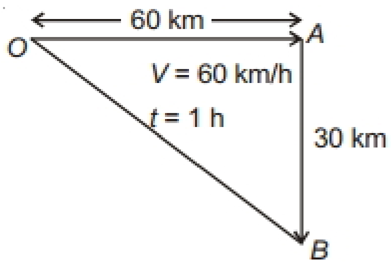
7. A car moves with speed $60\ km/h$ for 1 hour in east direction and with same speed for 30 min in south direction. The displacement of car from initial position is .... km

A) 60 B) $30\sqrt{3}$ C) $30\sqrt{5}$ D) $60\sqrt{2}$

Solution : (Correct Answer: C)

Displacement of car $= \sqrt{60^2 + 30^2} = 30\sqrt{5}\, km$

$\left[\begin{array}{l} \text{Distance } OA = \text{ Speed } \times \text{ Time} \\ \Rightarrow 60 \times 1h = 60\, km \end{array}\right]$



8. The numerical ratio of displacement to the distance covered is always

A) Less than one B) Equal to one

C) Equal to or less than one D) Equal to or greater than one

Solution : (Correct Answer: C)

(c)Since displacement is always less than or equal to distance, but never greater than distance. Hence numerical ratio of displacement to the distance covered is always equal to or less than one.

9. Which of the following is a one dimensional motion

A) Landing of an aircraft B) Earth revolving a round the sun

C) Motion of wheels of a moving trains D) Train running on a straight track

Solution : (Correct Answer: D)

Motion of train on straight track is a one dimensional motion whereas everything else is a two dimensional or three dimensional activity.

10. A body is moving from rest under constant acceleration and let S_1 be the displacement in the first $(p-1)$ sec and S_2 be the displacement in the first p sec. The displacement in $(p^2 - p + 1)^{th}$ sec. will be

A) $S_1 + S_2$ B) $S_1 S_2$ C) $S_1 - S_2$ D) S_1/S_2

Solution : (Correct Answer: A)

(a) From $S = ut + \frac{1}{2}a\, t^2$

$S_1 = \frac{1}{2}a(P-1)^2$ and $S_2 = \frac{1}{2}a\, P^2 \qquad [As\ u = 0]$

From $S_n = u + \frac{a}{2}(2n-1)$

$S_{(P^2-P+1)^{th}} = \frac{a}{2}\left[2(P^2 - P + 1) - 1\right]$

$= \frac{a}{2}\left[2P^2 - 2P + 1\right]$

It is clear that $S_{(P^2-P+1)^{th}} = S_1 + S_2$

11. The displacement of a body is given to be proportional to the cube of time elapsed. The magnitude of the acceleration of the body is

A) Increasing with time **B)** Decreasing with time

C) Constant but not zero **D)** Zero

Solution : (Correct Answer: A)

(a) $S = kt^3$

$\therefore a = \frac{d^2S}{dt^2} = 6kt$ i.e. $a \propto t$

12. Speed of two identical cars are u and $4u$ at a specific instant. The ratio of the respective distances in which the two cars are stopped from that instant is

A) $1:1$ **B)** $1:4$ **C)** $1:8$ **D)** $1:16$

Solution : (Correct Answer: D)

(d) $S \propto u^2 \Rightarrow \frac{S_1}{S_2} = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$

13. An athlete completes one round of a circular track of radius R in $40\,sec$. What will be his displacement at the end of $2\,min.\ 20\,sec$

A) $Zero$ **B)** $2R$ **C)** $2\pi R$ **D)** $7\pi R$

Solution : (Correct Answer: B)

(b)Total time of motion is $2\,min\,20\,sec = 140\,sec.$

As time period of circular motion is $40\,sec$ so in $140\,sec.$

Athlete will complete 3.5 revolution

i.e., He will be at diametrically opposite point i.e., Displacement $= 2R.$